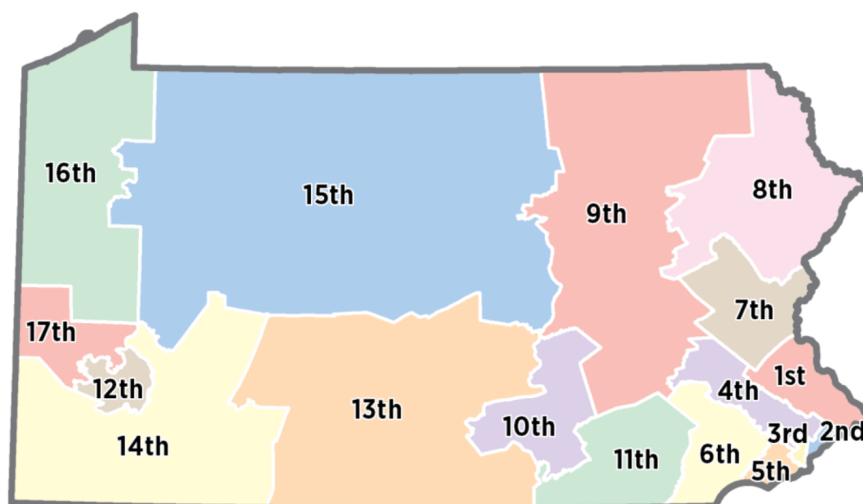


Final Project

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Code link: https://github.com/VivianBelle17/AIR_FINAL

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1 Introduction

This project analyzes congressional redistricting in Pennsylvania. The goal is to examine whether Pennsylvania's current congressional district map shows signs of partisan advantage or possible gerrymandering. Redistricting is the process of redrawing district boundaries after the Census so that districts have roughly equal population. In Pennsylvania, Congressional redistricting is not handled by the *Legislative Reapportionment Commission*, which redraws districts for the state Senate and House chambers; rather, the boundaries of Congressional seats are redrawn after every Federal decennial census by legislative action. This means that a bill first proceeds through both chambers of the

General Assembly and then is signed into law by the Governor. Pennsylvania has a total population of 13,002,700, with each congressional district averaging 764,865 people.

With that, gerrymandering occurs when district lines are drawn to benefit one political party or group, which Pennsylvania has a history of [1]. In 2011, for example, the state was successfully sued for its 2011 congressional map, in which the League of Women Voters of Pennsylvania argued that Pennsylvania’s elected officials manipulated the congressional district boundaries to ensure a majority Republican delegation in Congress and to minimize the ability of Democratic voters to elect U.S. House representatives; in which it was referred to as “one of the top three starkest partisan gerrymanders in the country and the worst in Pennsylvania’s history.” In this congressional map, Pennsylvania’s 7th district was infamously known as the “Goofy Kicking Donald Duck” [2] because of how distorted the boundaries appeared, resulting in this case becoming a major example of concerns surrounding partisan redistricting and unfair representation.

Furthermore, as Pennsylvania is a swing state, it is crucial to the outcome of General elections and, therefore, is a useful state to study due to its history of redistricting litigation and competitive elections. In this report, we use precinct-level election data, the 2020 Census population data, and the adopted 2022 congressional district plan, drawn by the state Supreme Court, to analyze and compare the results of various elections in the 2024 General election.

2 Methods

2.1 Data Collection and Preparation

The data for this project came from the Redistricting Data Hub [3]. We collected the draft of the Pennsylvania 2024 general election precinct data, 2020 Census block-level PL 94-171 population data, and the adopted 2022 congressional district shapefile, which served as the enacted map that we analyzed throughout the project. For election results, we used a draft version of the 2024 Pennsylvania General Election dataset, provided directly from the Redistricting Data Hub team, which included multiple statewide elections, such as the Presidential, Senate, and Attorney General races. The election data was then loaded as precinct geometries, while the population data was loaded at the census block level. The geometry repair process was computationally expensive, with some Smart Repair operations taking over eight hours to complete before producing a usable repaired shapefile. This highlights the complexity of repairing large precinct-level geographic datasets while preserving valid district geometry relationships. The final dataset retained geometry information, total population values (TOTPOP), racial demographic variables, congressional district assignments, and selected statewide election result columns required for GerryChain updaters and ensemble analysis.

To prepare the data before running our ensemble analysis, we cleaned and repaired the raw precinct geometry data using GeoPandas and MAUP. Initially, the data set had invalid geometries and overlaps, which actually prevented us from running `maup.doctor()`

Demographic Group	Population Lost
Hispanic	11
Asian	5
Black	4
American Indian	3
White	2
Other	1

Table 1: Population Differences After Repair

first. As a result, we first ran `maup.smartrepair()` with its default settings to repair the precinct geometries for further validation. After that, we ran `smartrepair` again using additional parameters, including nesting precincts within counties and fixing small rook adjacencies. This helped reduce overlaps and improve the overall geometry consistency of the dataset. We then assigned census blocks to precincts using `maup.assign()` to ensure that population and demographic data are properly transferred to the precinct-level dataset and assign each precinct to a congressional district. To validate the assignment process, we compared the total population before and after the assignment. The totals were nearly identical, with only a very small difference after repair. The original dataset contained 13,002,700 people, while the repaired dataset contained 13,002,674, resulting in a difference of 26 individuals; with the demographic differences summarized in Table 1. We decided to move forward with this election dataset due to this loss in population not having a strong effect on the analysis that will be discussed throughout the remainder of the analysis.

Finally, we used `maup.doctor()` to validate the repaired geometries. The final dataset includes precinct geometries, population totals, racial demographic columns, election results, and congressional district assignments. Although the final output still reported two remaining holes, the repaired dataset was still considered acceptable for analysis because the remaining geometry issues were extremely small relative to the statewide dataset and did not meaningfully affect population assignment, district connectivity, or election outcome calculations.

2.2 Ensemble Analysis

After cleaning the data, we used `GerryChain` to perform an ensemble analysis using a Markov Chain process. A Markov Chain works by repeatedly generating new districting plans from previous valid plans while following a set of constraints. Instead of evaluating only one map, this allows us to generate thousands of alternative maps and compare them to the enacted plan. To generate new plans, we used `ReCom`. This works by merging two neighboring districts together and then splitting them again in a different way, while still keeping both of the districts' population balanced. This creates realistic alternative districting plans while preserving contiguity. For our parameters, we used a

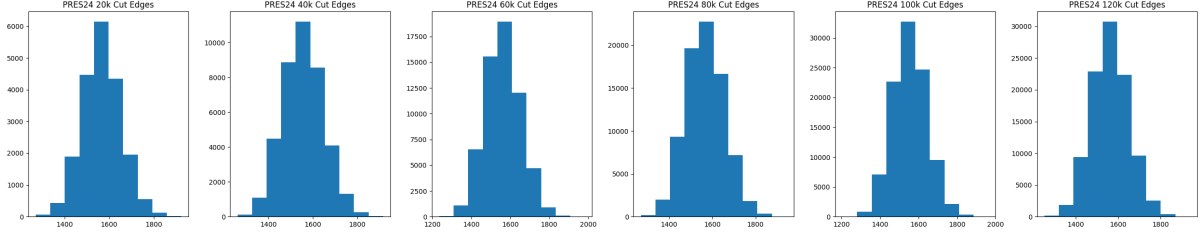


Figure 1: Presidential Election: Cut Edge Distribution Across All Chain Lengths

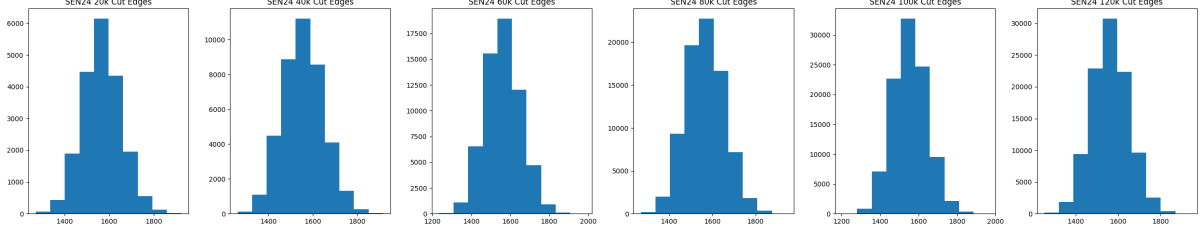


Figure 2: Senate Election: Cut Edge Distribution Across All Chain Lengths

population tolerance of 3% to ensure districts remained close to equal population. We analyzed Pennsylvania’s 17 congressional districts and used a `node_repeats` value of 2 during recombination. The `node_repeats` parameter controls how many attempts ReCom can make when searching for a valid balanced district split. Initially, lower values occasionally produced unsuccessful or less stable splits, while higher values increased runtime significantly. We ultimately selected a value of 2 because it improved the reliability of balanced splits without substantially slowing the Markov chain.

To study convergence, we ran six separate chains consisting of 20,000, 40,000, 60,000, 80,000, 100,000 steps and 120,000 steps. Comparing these chains helped us determine whether the ensemble distributions stabilized over time. We used these chains on three statewide elections, including the 2024 Presidential, Senate, and Attorney General races.

The cut-edge distributions across the Presidential, Senate, and Attorney General elections remained centered around similar ranges and maintained comparable overall shapes across increasing chain lengths, as shown in Figures 1, 2, and 3. This suggests that the Markov chain stabilized over time and that the ensemble behavior was driven more by the structural and geometric constraints of the districting process rather than the specific election dataset being analyzed. Cut edges are useful because they measure how frequently district boundaries split neighboring geographic units, providing insight into the structural complexity and compactness of generated districting plans.

Since cut edges measure the structural properties of district boundaries rather than election outcomes directly, this suggests that Markov chain behavior was influenced more by the geometric and population constraints of the districting process than by the specific election dataset. The consistency across elections also indicates stable sampling behavior within the chain.

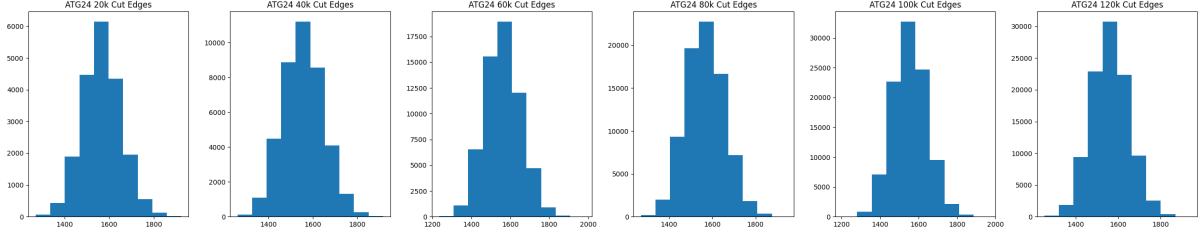


Figure 3: Attorney General Election: Cut Edge Distribution Across All Chain Lengths

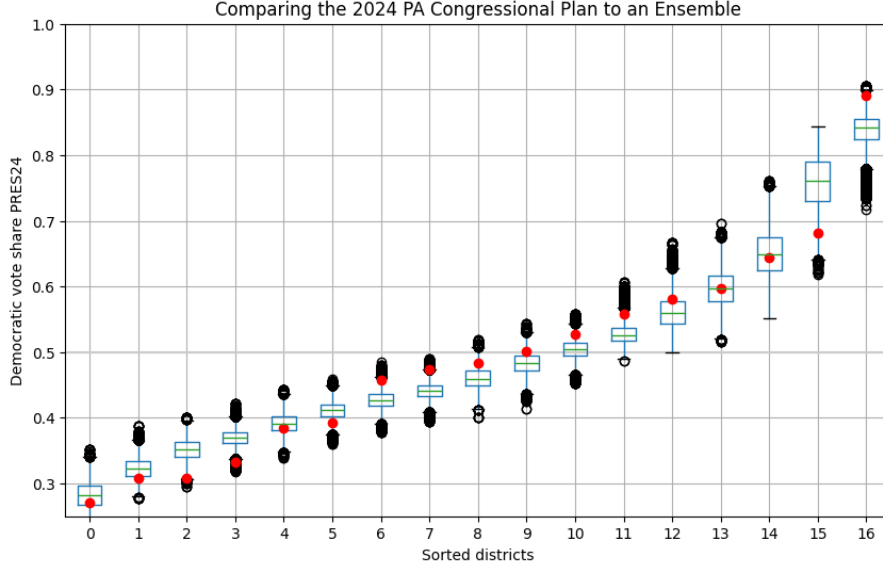


Figure 4: Presidential Election: Marginal Box Plot

2.3 Marginal Box Plots

This marginal box plot compares the enacted 2024 Pennsylvania congressional plan, shown by the red dots, against the ensemble of districting plans generated by our Markov Chain analysis. Each box plot represents the distribution of Democratic vote share across all generated plans. The districts shown in the graph are ordered from the least Democratic to the most Democratic. The graph shows an enacted Pennsylvania congressional plan (red dots) with the ensemble of randomly generated districting plans (the box plots). If the Red Dot Is Inside the Box, the enacted map looks typical/normal compared to the ensemble. If the Red Dot is near the edge, then the enacted map is somewhat unusual but still plausible. And if the Red Dot is outside of the whiskers, the enacted map may be an outlier/extreme compared to the ensemble.

Here are a few things to observe:

1. Most of the enacted districts fall within the ensemble distributions, which suggests that the enacted plan is generally behaving similarly to many valid alternative plans

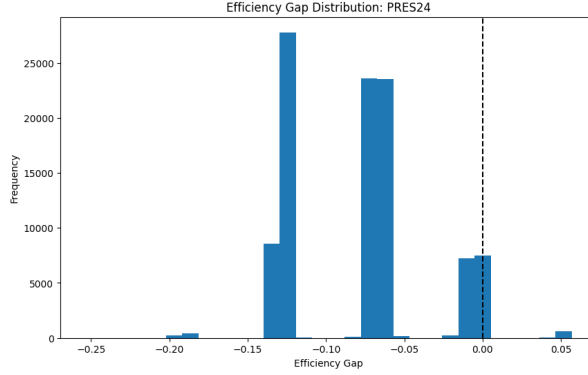


Figure 5: Presidential Election: Efficiency Gap

generated by the Markov Chain.

2. Some districts appear near the edges of the distributions. A few red dots are near the top whiskers or outside the tighter portions of the box plots. This may suggest that certain districts are more extreme compared to the ensemble.
3. The largest spread appears in highly Democratic districts. Toward the right side of the graph, the distributions become wider, especially in districts 15 and 16, which is evidence of variability in heavily Democratic districts across the ensemble plans.
4. There is no dramatic statewide outlier pattern. Because most red dots remain close to the ensemble distributions, the enacted plan does not appear to be an extreme statewide outlier based on Democratic vote share distributions alone.

3 Results

3.1 Efficiency Gap

The efficiency gap is a metric used to measure partisan advantage by comparing the number of “wasted votes” between political parties across districts. Figure 5 shows the distribution of efficiency gap values across the ensemble for the 2024 presidential election. The distribution shown looks more clustered than smooth, but that does not necessarily mean the graph is wrong. It likely means that many of the generated plans produced very similar partisan outcomes. Since Pennsylvania only has 17 congressional districts, there are limited ways for seats to shift between parties. As a result, the efficiency gap values can often repeat, creating these tall spikes in the histogram.

Therefore, our interpretation is that the ensemble produces different maps, but many of the maps result in similar election outcomes. This suggests that Pennsylvania’s partisan results may be relatively stable under the constraints we used in the Markov chain. The dashed vertical line at zero represents partisan balance, where neither party has an

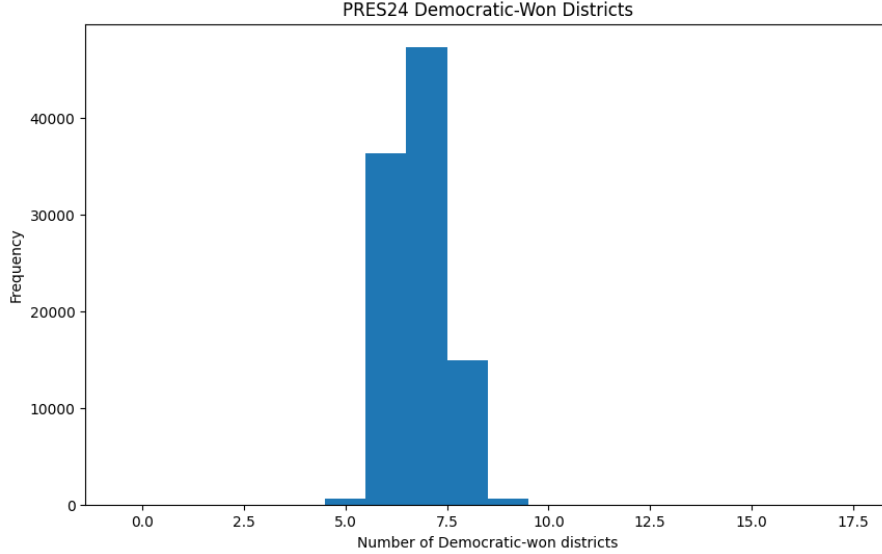


Figure 6: Presidential Election: Democratic Won Districts

advantage. Values to the left of zero indicate a Democratic advantage, while values to the right indicate a Republican advantage. Most of the generated plans fall on the negative side of the graph, specifically around -0.13 and -0.06, suggesting that many valid districting plans in the ensemble slightly favor Democrats under the presidential election results.

We can also see multiple clusters in the distribution rather than one smooth peak. This suggests that different district configurations can produce different levels of partisan advantage even while satisfying the same redistricting constraints. Only a small number of plans fall on the positive side of the graph, meaning Republican-favoring outcomes were relatively uncommon in these ensembles.

The repeated clustering patterns across the efficiency gap distributions suggest that many alternative districting plans converge toward similar partisan outcomes despite differences in district boundaries. This indicates that Pennsylvania’s political geography itself may naturally constrain the range of possible partisan outcomes under valid redistricting plans.

3.2 Democratic Won Districts

The histogram in Figure 6 shows the number of Democratic-won districts across the ensemble with data from the 2024 Presidential election. Most of the generated plans produce between 6 and 8 Democratic-won districts, with the highest concentration around 7 districts. This means that the most common outcome across the ensemble is for Democrats to win approximately seven seats. The distribution is relatively narrow, indicating that many valid districting plans produce fairly similar partisan seat out-

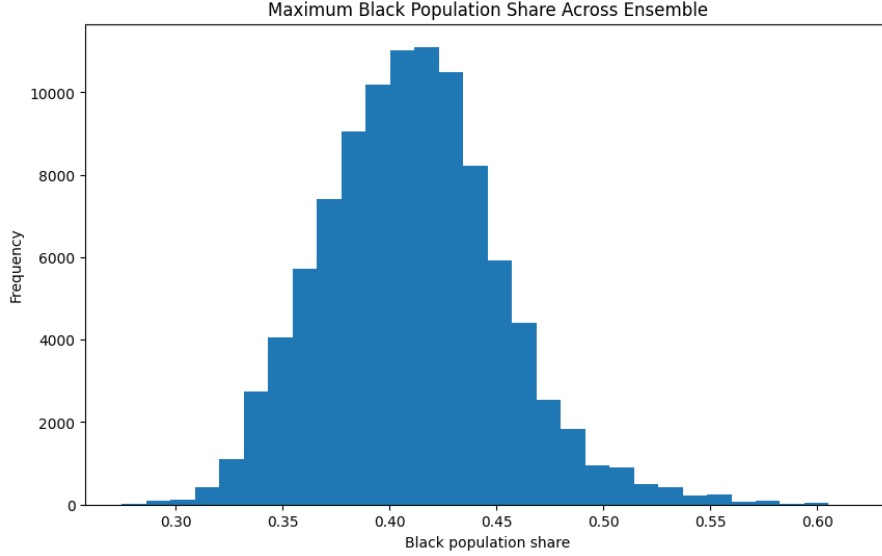


Figure 7: Maximum Black Population Share Across Ensemble

comes. Only a small number of plans fall outside this range, such as plans that result in Democrats winning 5 or 9 districts. Overall, this indicates that the political geography and districting constraints used in the ensemble tend to produce Democratic seat outcomes within this range.

3.3 Demographics

Figure 7 shows the maximum share of the Black population in any district between all plans in the ensemble. Most of the generated plans produce a district with roughly 40% Black population, which is where the distribution is most concentrated. We can also see a slight right skew, meaning that some plans are capable of producing districts with higher Black population shares, although those become less common as the percentage increases. Very few plans create districts with a Black population above 50%, which suggests that majority-Black districts are relatively uncommon under the constraints and geography used in this ensemble.

Figure 8 shows the maximum Hispanic population shares in the ensemble plans. Unlike the distribution of the Black population, the shares of the Hispanic population are much lower, mostly between about 19% and 23%. The distribution is also narrower, showing that there is less variation between the plans. This suggests that most valid districting plans naturally produce fairly similar Hispanic population concentrations, while a high Hispanic population in districts is uncommon in the ensemble.

For our next analysis, we decided to examine how many majority-Black and majority-Hispanic districts appeared in the ensemble of generated plans. These histograms show that almost all generated plans contained zero majority-Black and zero majority-Hispanic

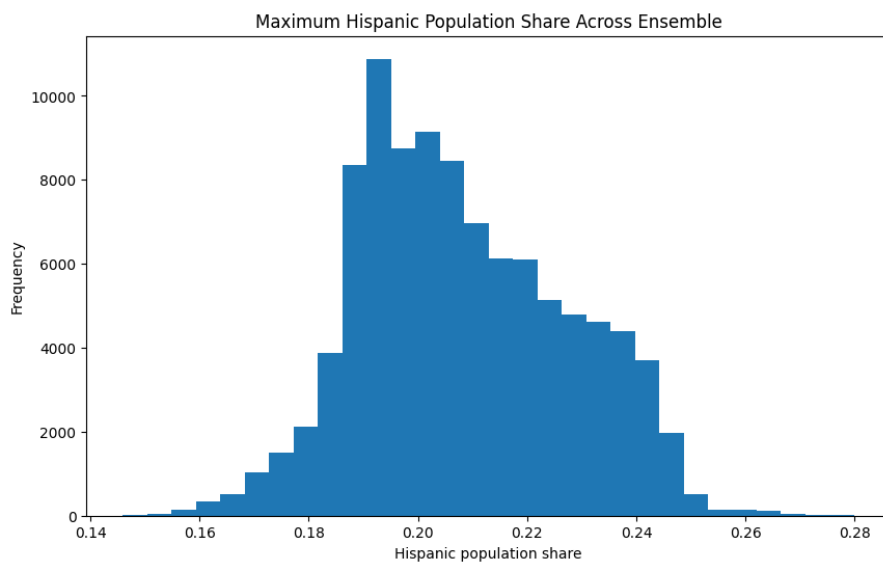


Figure 8: Maximum Hispanic Population Share Across Ensemble

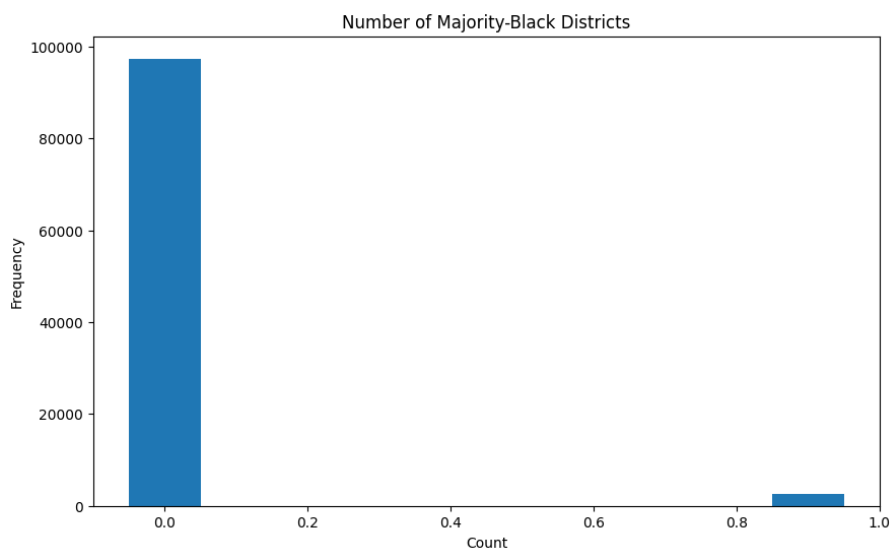


Figure 9: Majority Black Districts

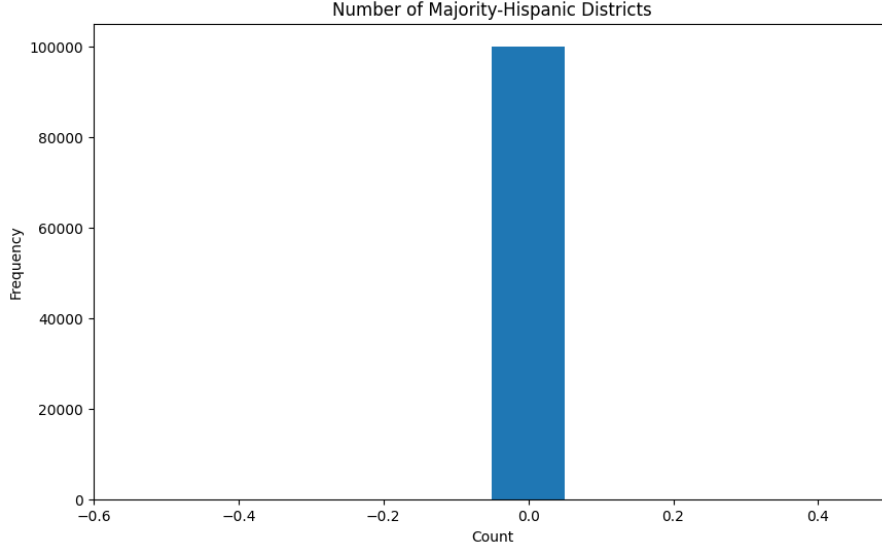


Figure 10: Majority Hispanic Districts

districts. In the majority-Black histogram, there are a small number of plans that contain one majority-Black district, but this occurs relatively infrequently compared to the overall ensemble. The majority-Hispanic histogram is even more concentrated at zero, suggesting that none of the generated plans produced a district where the Hispanic population exceeded 50%. Overall, these results suggest that, under the constraints of the ensemble and the geographic distribution of the population, majority-minority districts were generally difficult to form in these plans. Because these results initially appeared unexpected, we compared the ensemble results to the enacted districting plan and it showed that there was only one black majority district and zero Hispanic majority districts. With that, our ensemble remained similar to that regard.

3.4 Comparing Multiple Elections

3.4.1 Efficiency Gap

Figures 11 and 12 show the efficiency gap distributions for the 2024 Attorney General and Senate elections across the ensemble plans. As mentioned before, the dashed vertical line at zero represents perfect partisan balance, values to the left indicate Democratic advantage, while values to the right indicate Republican advantage. In both elections, most of the generated plans fall on the negative side of the graph, meaning that many valid districting plans slightly favor Democrats under these election results.

There are also several distinct clusters in both distributions rather than one smooth peak. This suggests that different district configurations within the ensemble can create different levels of partisan advantage while still satisfying the same districting constraints. The Attorney General distribution appears farther left overall, suggesting this election

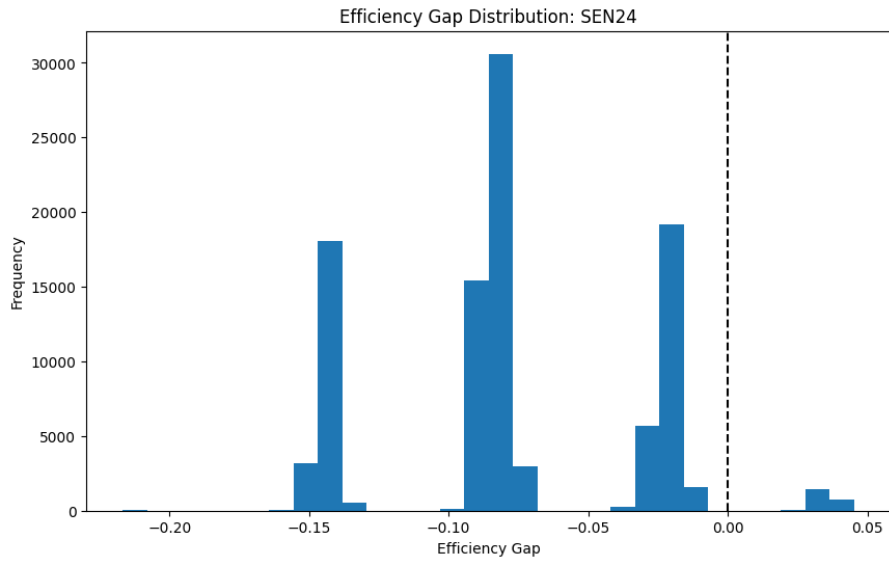


Figure 11: Senate Election: Efficiency Gap

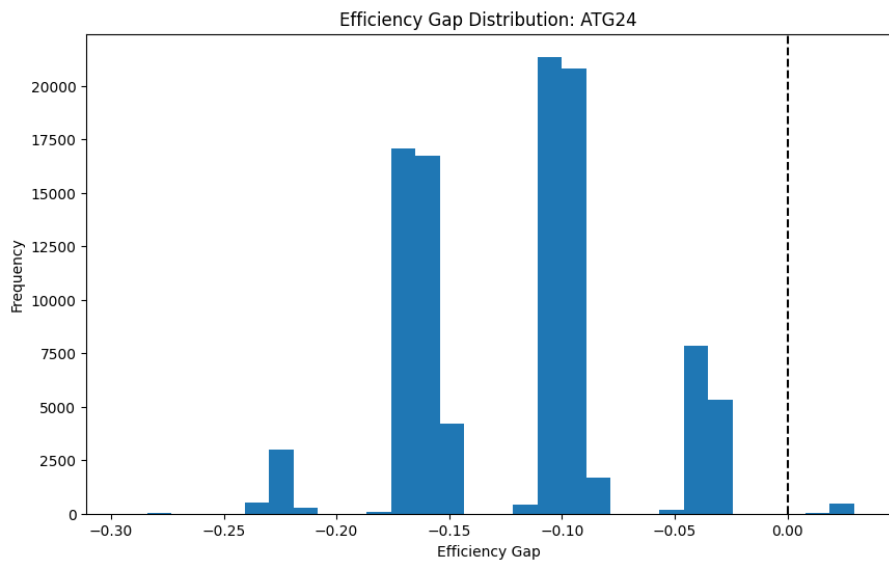


Figure 12: Attorney General: Efficiency Gap

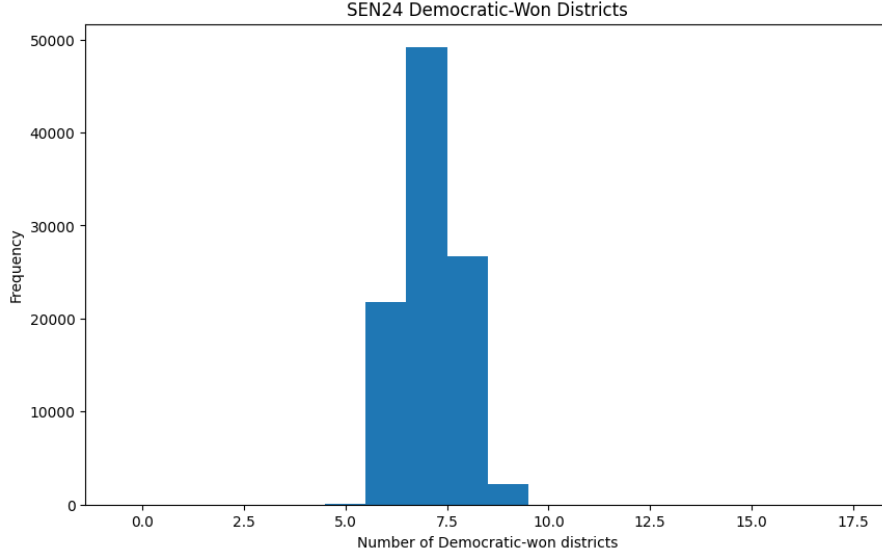


Figure 13: Senate Election: Democratic Won Districts

had somewhat stronger Democratic advantages compared to the Senate distribution, while the Senate results include some clusters closer to zero that represent more balanced outcomes. Overall, both elections show fairly consistent patterns with relatively few Republican-favoring plans appearing in the ensemble.

3.4.2 Democratic Won Districts

Figures 13 and 14 show the number of Democratic-won districts across the ensemble plans for the 2024 Attorney General and Senate elections.

In both elections, most generated plans produce between about 5 and 7 Democratic-won districts, with the highest concentration centered around 6 to 7 districts. Indicating the ensemble tends to produce fairly consistent partisan seat outcomes across both statewide races. The Senate distribution appears slightly more concentrated around 7 Democratic seats, while the Attorney General distribution shows a little more variation toward 5 and 6 seats. However, the overall patterns remain very similar between the two elections. The relatively narrow distributions suggest that many valid districting plans produce comparable election outcomes under the ensemble constraints, and only a small number of plans fall outside the main range. Overall, the similarity between these graphs suggests that the ensemble behavior remains stable across different statewide elections rather than being driven by a single race.

3.4.3 Marginal Box Plots

To determine whether these patterns in the presidential graph were consistent across elections, we repeated the same analysis using the 2024 Senate and Attorney General

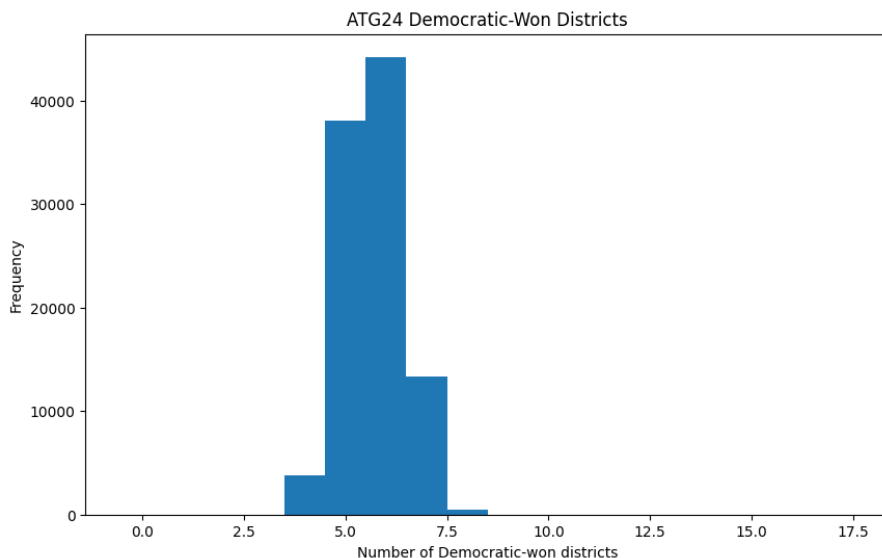


Figure 14: Attorney General Election: Democratic Won Districts

racers.

Figure 15 shows the same marginal box plot analysis using the 2024 Senate election data. Similar to the presidential results, most enacted districts remain within the ensemble distributions, suggesting that the enacted Pennsylvania map behaves similarly to many valid alternative plans generated by the ensemble. We also continue to see greater variability in the more heavily Democratic districts toward the right side of the graph, while the more Republican-leaning districts remain relatively stable. Overall, the Senate results follow a pattern very similar to the presidential election, which suggests that the ensemble behavior remains consistent across different statewide elections.

Figure 16 shows the marginal box plot analysis for the 2024 Attorney General election. Again, the enacted districts largely fall within the ensemble distributions, and the overall district ordering remains very similar to the Presidential and Senate elections. The consistency across all three elections suggests that the patterns we observed are not isolated to a single race, but instead reflect broader structural behavior within the districting plans generated by the ensemble. Overall, these results strengthen the conclusion that the enacted plan is not a dramatic statewide outlier based on Democratic vote share distributions.”

4 Discussion/Conclusions

The main takeaway from our project is that the Markov chain produced fairly stable results across all three elections: Presidential, Senate, and Attorney General. Since the cut-edge distributions and election results looked similar, this suggests that the chain was behaving consistently. We also found that the enacted Pennsylvania congressional map

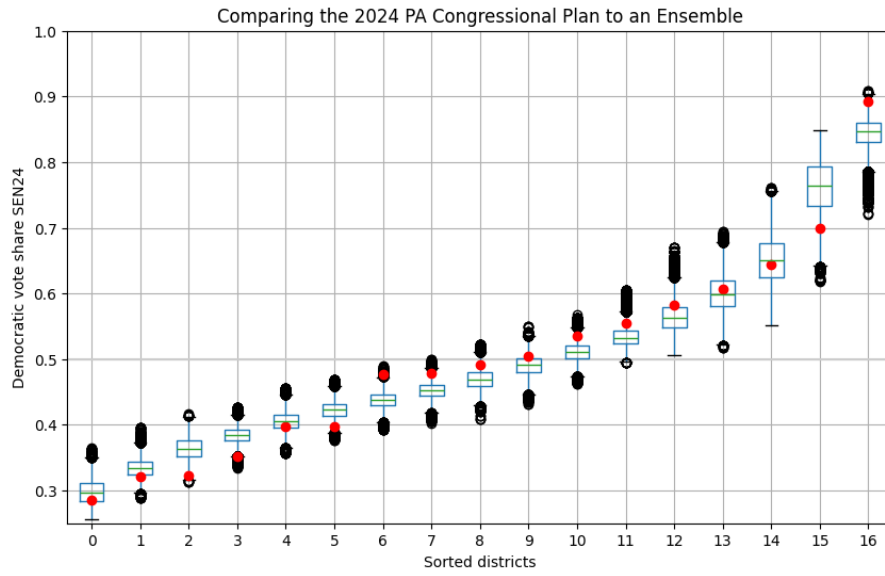


Figure 15: Senate Election: Marginal Box Plot

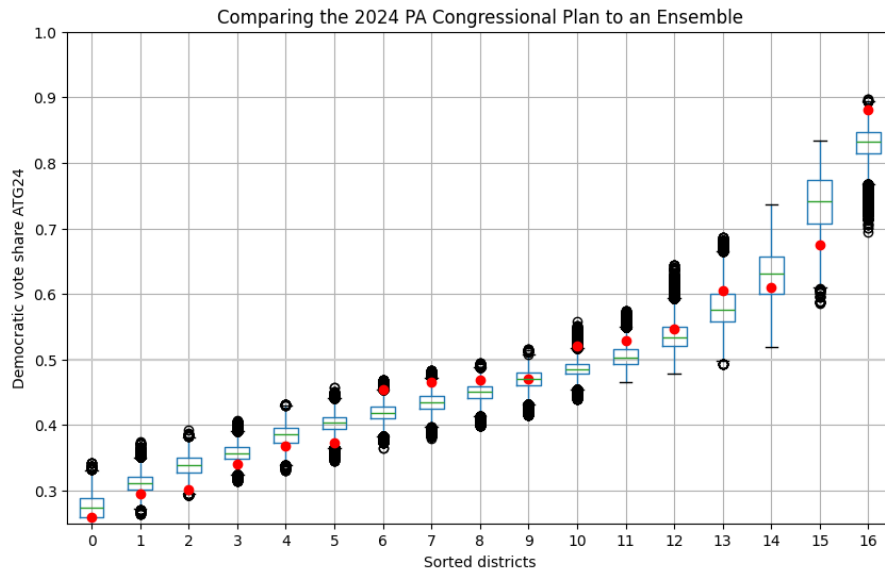


Figure 16: Attorney General Election: Marginal Box Plot

generally fit within the ensemble of alternative plans. This means the current map did not appear to be an extreme outlier compared to the maps generated by our chain. Smart Repair caused a small population loss of 26 people; this was extremely small compared to Pennsylvania’s total population of over 13 million. Because of that, we do not think it meaningfully affected our final results, but we still documented it as a limitation.

From our demographic analysis, we found that representation opportunities appeared to be strongly influenced by Pennsylvania’s underlying population geography. The “Majority Black” and “Majority Hispanic” ensemble results showed that most plans only produced about one majority-Black district, while almost no plans produced a majority-Hispanic district. This suggests that the geographic distribution of Pennsylvania’s population may naturally limit the creation of additional majority-minority congressional districts under the constraints used in the ensemble.

References

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